Dao T. Bui

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Education

University of South Carolina, Honors College — Columbia, South Carolina
B.S. Computer Science (Artificial Intelligence Concentration) — *May 2029*

- **GPA:** 4.0 / 4.0
- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Discrete Structures, Computer Architecture, Unix & Linux
- **Scholarships:** Palmetto Fellows ($40,000), Dean's Scholarship ($6,000), Presidential Scholarship ($2,000)

Technical Skills

Languages: C++, Python, Java, JavaScript/TypeScript, HTML/CSS
Frameworks & Libraries: React/Next.js, FastAPI, PyTorch, OpenCV, MediaPipe, Tailwind, Chart.js, NumPy, pandas
Systems & Tools: Linux/Unix, Bash, Git/GitHub, Docker, REST/HTTP, Jupyter, LaTeX

Software Engineering Experience

Adaptive Real-Time Systems Lab, University of South Carolina — Columbia, South Carolina
Machine Learning Research Assistant — *Jun 2025 – Present*

- Implemented a real-time spatter detection pipeline with **YOLOv8** and **OpenCV**; trained with **PyTorch** on Linux/HPC to improve accuracy and latency.
- Annotated 200+ images in **Label Studio** and optimized dataloaders/augmentations to speed up training and inference.
- Drafting first-author submission (SPIE) and planning deployment on a **Raspberry Pi** for on-device inference and monitoring.

Brain Cognition Lab, Medical University of South Carolina — Charleston, South Carolina
Research Intern — *Jun 2024 – Jul 2024*

- Analyzed cognitive-test datasets (N $\approx$ 220) to identify performance trends and early-detection indicators for Alzheimer's patients.
- Developed Python scripts to preprocess brain-scan data (denoising, normalization) and verify scan quality.
- Presented findings on facial recognition and episodic memory at the SC ADRC Symposium.

Projects

Computer Vision Sobriety Test | *Next.js/TypeScript, FastAPI (Python), MediaPipe, OpenCV* — Sep 2025

- Designed a real-time pose/gaze pipeline (30 FPS) using **MediaPipe/OpenCV**; exposed REST endpoints via **FastAPI**.
- Built a responsive **Next.js** frontend and integrated a Python inference service; added metrics logging for latency and accuracy (>85%).

NASA Hacks — SAR Visualization Platform | *Full-stack app, SAR data, open-source maps* — 2025

- Won **Best Use of Satellite Data** at **NASA Hacks** (25+ teams) by analyzing **SAR** datasets to visualize global oil spill impact.
- Built a full-stack app to ingest/process SAR data and render an interactive UI synced with an open-source mapping library. GitHub

Full Stack Portfolio Backtester | *FastAPI, Python, Tailwind, Chart.js* — Jul 2025 – Sep 2025

- Implemented REST APIs to backtest **50+** assets across **1,200+** trading days; typical request completes in <3s via vectorization/caching.
- Built interactive charts (Chart.js) and analytics (Sharpe, volatility, 60/40) with structured JSON for the UI layer.

Awards

NASA Hacks — Best Use of Satellite Data (25+ teams) — 2025

Leadership

Hackathon Committee — University of South Carolina — Sep 2025 – Present

- Co-founding UofSC's first hackathon; secured **2 sponsors** (Capgemini, Dominion Energy) and led cross-campus promotion.